

Surface Model

5.4

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Module Index

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5.1 File List

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Module Documentation

6.1 Models

Modules

- [Utils](#)

6.1.1 Detailed Description

6.2 Utils

Modules

- [SurfaceModel](#)

6.2.1 Detailed Description

6.3 SurfaceModel

Files

- file [class_declarations.hh](#)
Forward declarations of classes defined for JEOD 2.0 surface model.
- file [cylinder.hh](#)
cylinders for use in the surface model and the contact model
- file [facet.hh](#)
Individual facets for use in the surface model.
- file [facet_params.hh](#)
A virtual base class for facet parameters, used to create interaction facets in the InteractionSurfaceFactorys.
- file [flat_plate.hh](#)
Flat plates for use in the surface model.
- file [flat_plate_circular.hh](#)

- circulat flat plates for use in the surface model and the contact model*
- file [flat_plate_thermal.hh](#)
Flat plates for use in the surface model, including a thermal portion.
- file [interaction_facet.hh](#)
Individual facets for use with specific environment interaction models.
- file [interaction_facet_factory.hh](#)
Factory that creates an interaction facet, for a specific environment interaction model, from a facet model.
- file [interaction_surface.hh](#)
Vehicle surface model for general environment interaction models.
- file [interaction_surface_factory.hh](#)
Factory that creates an interaction surface, for a specific environment interaction model, from a surface model.
- file [surface_model.hh](#)
Vehicle surface model for general environment interaction models.
- file [surface_model_messages.hh](#)
Implement surface_model_messages.
- file [facet.cc](#)
Individual facets for use in the surface model.
- file [flat_plate.cc](#)
Flat plates for use in the surface model.
- file [interaction_surface_factory.cc](#)
Factory that creates an interaction surface, for a specific enviornment interaction model, from a surface model.
- file [surface_model.cc](#)
Vehicle surface model for general environment interaction models.
- file [surface_model_messages.cc](#)
Implement surface_model_messages.

6.3.1 Detailed Description

Chapter 7

Namespace Documentation

7.1 jeod Namespace Reference

Namespace jeod.

Data Structures

- class [Cylinder](#)
An cylinder implementation of [Facet](#).
- class [Facet](#)
A general base class for all surface model facets.
- class [FacetParams](#)
General base class for all parameters associated with facets in the surface model.
- class [FlatPlate](#)
A [FlatPlate](#) implementation of [Facet](#).
- class [FlatPlateCircular](#)
An circular flat plate implementation of [Facet](#).
- class [FlatPlateThermal](#)
A [FlatPlate](#) implementation of [Facet](#), with thermal information.
- class [InteractionFacet](#)
A base class for an interaction specific facet.
- class [InteractionFacetFactory](#)
A factory to create a specific interaction facet from a general facet.
- class [InteractionSurface](#)
A base class for interaction specific surfaces.
- class [InteractionSurfaceFactory](#)
A base class for creating specific interaction surfaces from general surfaces.
- class [FacetStateInfo](#)
This is a structure used only in the surface model to aid in relative state calculations for articulation.
- class [SurfaceModel](#)
A general, non-interaction specific surface that can be used to create surfaces suitable for specific interactions.
- class [SurfaceModelMessages](#)
Messages associated with the use of the surface model.

7.1.1 Detailed Description

Namespace jeod.

Chapter 8

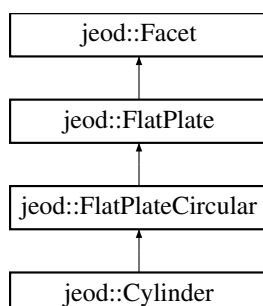
Data Structure Documentation

8.1 jeod::Cylinder Class Reference

An cylinder implementation of [Facet](#).

```
#include <cylinder.hh>
```

Inheritance diagram for jeod::Cylinder:



Public Member Functions

- [Cylinder](#) ()=default
- [~Cylinder](#) () override=default
- [Cylinder](#) & [operator=](#) (const [Cylinder](#) &)=delete
- [Cylinder](#) (const [Cylinder](#) &)=delete

Data Fields

- double [length](#) {}
Length of the cylinder.

Private Member Functions

- JEOD_CLASS_ESTABLISH_FRIENDS2 p (jeod, [Cylinder](#))

Additional Inherited Members

8.1.1 Detailed Description

An cylinder implementation of [Facet](#).

Definition at line 74 of file cylinder.hh.

8.1.2 Constructor & Destructor Documentation

8.1.2.1 Cylinder() [1/2]

```
jeod::Cylinder::Cylinder ( ) [default]
```

8.1.2.2 ~Cylinder()

```
jeod::Cylinder::~~Cylinder ( ) [override], [default]
```

8.1.2.3 Cylinder() [2/2]

```
jeod::Cylinder::Cylinder (
    const Cylinder & ) [delete]
```

8.1.3 Member Function Documentation

8.1.3.1 operator=()

```
Cylinder& jeod::Cylinder::operator= (
    const Cylinder & ) [delete]
```

8.1.3.2 p()

```
JEOD_CLASS_ESTABLISH_FRIENDS2 jeod::Cylinder::p (
    jeod ,
    Cylinder ) [private]
```


8.1.4 Field Documentation

8.1.4.1 length

```
double jeod::Cylinder::length {}
```

Length of the cylinder.

trick_units(m)

Definition at line 85 of file cylinder.hh.

The documentation for this class was generated from the following file:

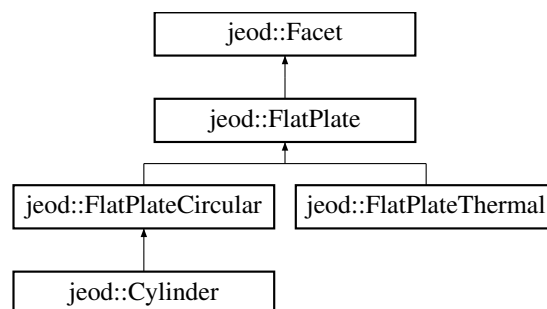
- [cylinder.hh](#)

8.2 jeod::Facet Class Reference

A general base class for all surface model facets.

```
#include <facet.hh>
```

Inheritance diagram for jeod::Facet:



Public Member Functions

- [Facet](#) ()=default
- virtual [~Facet](#) ()=default
- [Facet](#) & [operator=](#) (const [Facet](#) &)=delete
- [Facet](#) (const [Facet](#) &)=delete
- virtual void [initialize_mass_connection](#) (BaseDynManager &manager)
- virtual void [update_articulation](#) ()
- MassBody * [get_mass_body_ptr](#) ()
- MassPointState * [get_mass_rel_struct](#) ()
 - Getter for the mass_rel_struct element,.*
- void [set_name](#) (std::string name_in)
 - Setter for the name.*

Data Fields

- double `position` [3] {}
Position of the facet in the vehicle structural frame.
- double `local_position` [3] {}
Position of the facet in the structure frame of the MassBody this facet is associated with.
- std::string `param_name`
Name of the facet parameters, usually a material type.
- std::string `name`
Name of the facet.
- std::string `mass_body_name`
The name of the MassBody this facet is associated with.
- double `temperature` {}
Kinetic Temperature of the surface.
- double `area` {}
Area of the plate.

Protected Member Functions

- virtual void `update_articulation_internal` ()

Protected Attributes

- MassBody * `mass_body_ptr` {}
A pointer to the MassBody this facet is associated with.
- MassPointState * `mass_rel_struct` {}
The relative state between the mass body this facet is associated with, and the user set structure point in the mass tree.
- double `int_pos` [3] {}
An intermediate position, used as a working variable when doing calculations.
- bool `connections_initialized` {}
Indicates if the mass connections for this Facet have been initialized.

Private Member Functions

- JEOD_CLASS_ESTABLISH_FRIENDS2 f (jeod, `Facet`)

Friends

- class `SurfaceModel`

8.2.1 Detailed Description

A general base class for all surface model facets.

Definition at line 88 of file facet.hh.

8.2.2 Constructor & Destructor Documentation

8.2.2.1 Facet() [1/2]

```
jeod::Facet::Facet ( ) [default]
```

8.2.2.2 ~Facet()

```
virtual jeod::Facet::~~Facet ( ) [virtual], [default]
```

8.2.2.3 Facet() [2/2]

```
jeod::Facet::Facet (
    const Facet & ) [delete]
```

8.2.3 Member Function Documentation

8.2.3.1 f()

```
JEOD_CLASS_ESTABLISH_FRIENDS2 jeod::Facet::f (
    jeod ,
    Facet ) [private]
```

8.2.3.2 get_mass_body_ptr()

```
MassBody * jeod::Facet::get_mass_body_ptr ( )
```

Definition at line 119 of file facet.cc.

References mass_body_ptr.

8.2.3.3 get_mass_rel_struct()

```
MassPointState* jeod::Facet::get_mass_rel_struct ( ) [inline]
```

Getter for the mass_rel_struct element,.

Definition at line 107 of file facet.hh.

8.2.3.4 initialize_mass_connection()

```
void jeod::Facet::initialize_mass_connection (
    BaseDynManager & manager ) [virtual]
```

Definition at line 57 of file facet.cc.

References connections_initialized, jeod::SurfaceModelMessages::initialization_error, mass_body_name, and mass_body_ptr.

8.2.3.5 operator=()

```
Facet& jeod::Facet::operator= (
    const Facet & ) [delete]
```

8.2.3.6 set_name()

```
void jeod::Facet::set_name (
    std::string name_in ) [inline]
```

Setter for the name.

Definition at line 115 of file facet.hh.

8.2.3.7 update_articulation()

```
void jeod::Facet::update_articulation ( ) [virtual]
```

Definition at line 87 of file facet.cc.

References connections_initialized, jeod::SurfaceModelMessages::initialization_error, and update_articulation_↔ internal().

8.2.3.8 update_articulation_internal()

```
void jeod::Facet::update_articulation_internal ( ) [protected], [virtual]
```

Reimplemented in [jeod::FlatPlate](#).

Definition at line 130 of file facet.cc.

References `int_pos`, `local_position`, `mass_rel_struct`, and `position`.

Referenced by `update_articulation()`, and `jeod::FlatPlate::update_articulation_internal()`.

8.2.4 Friends And Related Function Documentation

8.2.4.1 SurfaceModel

```
friend class SurfaceModel [friend]
```

Definition at line 90 of file facet.hh.

8.2.5 Field Documentation

8.2.5.1 area

```
double jeod::Facet::area {}
```

Area of the plate.

`trick_units(m2)`

Definition at line 163 of file facet.hh.

8.2.5.2 connections_initialized

```
bool jeod::Facet::connections_initialized {} [protected]
```

Indicates if the mass connections for this [Facet](#) have been initialized.

`trick_units(-)`

Definition at line 193 of file facet.hh.

Referenced by `initialize_mass_connection()`, and `update_articulation()`.

8.2.5.3 int_pos

```
double jeod::Facet::int_pos[3] {} [protected]
```

An intermediate position, used as a working variable when doing calculations.

trick_units(m)

Definition at line 187 of file facet.hh.

Referenced by update_articulation_internal().

8.2.5.4 local_position

```
double jeod::Facet::local_position[3] {}
```

Position of the facet in the structure frame of the MassBody this facet is associated with.

Used in articulation, contact, etc. trick_units(m)

Definition at line 133 of file facet.hh.

Referenced by update_articulation_internal().

8.2.5.5 mass_body_name

```
std::string jeod::Facet::mass_body_name
```

The name of the MassBody this facet is associated with.

This is used, only for specific applications, such as contact and articulation. Otherwise it is optional. This is used to find and cache a pointer to the mass_body, stored below in 'mass_body_ptr'. trick_units(-)

Definition at line 153 of file facet.hh.

Referenced by initialize_mass_connection().

8.2.5.6 mass_body_ptr

```
MassBody* jeod::Facet::mass_body_ptr {} [protected]
```

A pointer to the MassBody this facet is associated with.

This is used, only for specific applications, such as contact and articulation. Otherwise it is optional. This pointer is cached from the tree of MassBodies for the mass_body_name set above trick_units(-)

Definition at line 174 of file facet.hh.

Referenced by get_mass_body_ptr(), and initialize_mass_connection().

8.2.5.7 mass_rel_struct

```
MassPointState* jeod::Facet::mass_rel_struct {} [protected]
```

The relative state between the mass body this facet is associated with, and the user set structure point in the mass tree.

Only used in certain applications, such as articulation. `trick_units(-)`

Definition at line 181 of file `facet.hh`.

Referenced by `update_articulation_internal()`, and `jeod::FlatPlate::update_articulation_internal()`.

8.2.5.8 name

```
std::string jeod::Facet::name
```

Name of the facet.

`trick_units(-)`

Definition at line 144 of file `facet.hh`.

8.2.5.9 param_name

```
std::string jeod::Facet::param_name
```

Name of the facet parameters, usually a material type.

Used to match the facet to [FacetParams](#) `trick_units(-)`

Definition at line 139 of file `facet.hh`.

8.2.5.10 position

```
double jeod::Facet::position[3] {}
```

Position of the facet in the vehicle structural frame.

The reference point of the facet is positioned on a class by class basis (example: for flat plate it is the position of the center of pressure) `trick_units(m)`

Definition at line 127 of file `facet.hh`.

Referenced by `update_articulation_internal()`.

8.2.5.11 temperature

```
double jeod::Facet::temperature {}
```

Kinetic Temperature of the surface.

trick_units(K)

Definition at line 158 of file facet.hh.

The documentation for this class was generated from the following files:

- [facet.hh](#)
- [facet.cc](#)

8.3 jeod::FacetParams Class Reference

General base class for all parameters associated with facets in the surface model.

```
#include <facet_params.hh>
```

Public Member Functions

- [FacetParams](#) ()=default
- virtual [~FacetParams](#) ()=default
- [FacetParams](#) & [operator=](#) (const [FacetParams](#) &)=delete
- [FacetParams](#) (const [FacetParams](#) &)=delete
- void [set_name](#) (std::string name_in)

Setter for the name.

Data Fields

- std::string [name](#)
- Name that will be used to match [FacetParams](#) to facets.*

Private Member Functions

- JEOD_CLASS_ESTABLISH_FRIENDS2 [p](#) (jeod, [FacetParams](#))

8.3.1 Detailed Description

General base class for all parameters associated with facets in the surface model.

Definition at line 80 of file facet_params.hh.

8.3.2 Constructor & Destructor Documentation

8.3.2.1 FacetParams() [1/2]

```
jeod::FacetParams::FacetParams ( ) [default]
```

8.3.2.2 ~FacetParams()

```
virtual jeod::FacetParams::~~FacetParams ( ) [virtual], [default]
```

8.3.2.3 FacetParams() [2/2]

```
jeod::FacetParams::FacetParams (
    const FacetParams & ) [delete]
```

References name.

8.3.3 Member Function Documentation

8.3.3.1 operator=()

```
FacetParams& jeod::FacetParams::operator= (
    const FacetParams & ) [delete]
```

8.3.3.2 p()

```
JEOD_CLASS_ESTABLISH_FRIENDS2 jeod::FacetParams::p (
    jeod ,
    FacetParams ) [private]
```

8.3.3.3 set_name()

```
void jeod::FacetParams::set_name (
    std::string name_in ) [inline]
```

Setter for the name.

Definition at line 96 of file facet_params.hh.

8.3.4 Field Documentation

8.3.4.1 name

```
std::string jeod::FacetParams::name
```

Name that will be used to match [FacetParams](#) to facets.

trick_units(-)

Definition at line 91 of file facet_params.hh.

Referenced by jeod::InteractionSurfaceFactory::add_facet_params(), and FacetParams().

The documentation for this class was generated from the following file:

- [facet_params.hh](#)

8.4 jeod::FacetStateInfo Class Reference

This is a structure used only in the surface model to aid in relative state calculations for articulation.

```
#include <surface_model.hh>
```

Public Member Functions

- [FacetStateInfo](#) ()=default
Default constructor to keep the memory manager happy.
- [FacetStateInfo](#) (MassBody *new_mass_body)
FacetStateInfo non-default constructor.
- [FacetStateInfo](#) (MassBody &new_mass_body)
FacetStateInfo non-default constructor.
- bool [operator==](#) (const [FacetStateInfo](#) &rhs) const
Compare this FacetStateInfo object to another.

Data Fields

- MassPointState [mass_state](#)
The resulting relative mass point state between the structural body named in struct_body_name and the MassBody pointed to in this structure's mass_body.
- MassBody * [mass_body](#) {}
The MassBody object whose state, relative and w.r.t.

Private Member Functions

- JEOD_CLASS_ESTABLISH_FRIENDS2 [p](#) (jeod, [FacetStateInfo](#))

8.4.1 Detailed Description

This is a structure used only in the surface model to aid in relative state calculations for articulation.

For each mass body that needs a relative state calculated w.r.t. the mass body named in struct_body_name, one of these objects will be instantiated. That way, the relative state information must only be calculated once per mass body.

Definition at line 95 of file surface_model.hh.

8.4.2 Constructor & Destructor Documentation

8.4.2.1 FacetStateInfo() [1/3]

```
jeod::FacetStateInfo::FacetStateInfo ( ) [default]
```

Default constructor to keep the memory manager happy.

8.4.2.2 FacetStateInfo() [2/3]

```
jeod::FacetStateInfo::FacetStateInfo (
    MassBody * new_mass_body ) [inline], [explicit]
```

[FacetStateInfo](#) non-default constructor.

Parameters

<i>new_mass_body</i>	The mass body to which this object will refer.
----------------------	--

Definition at line 120 of file surface_model.hh.

8.4.2.3 FacetStateInfo() [3/3]

```
jeod::FacetStateInfo::FacetStateInfo (
    MassBody & new_mass_body ) [inline], [explicit]
```

[FacetStateInfo](#) non-default constructor.

Parameters

<i>new_mass_body</i>	The mass body to which this object will refer.
----------------------	--

Definition at line 129 of file `surface_model.hh`.

8.4.3 Member Function Documentation

8.4.3.1 operator==()

```
bool jeod::FacetStateInfo::operator== (
    const FacetStateInfo & rhs ) const [inline]
```

Compare this [FacetStateInfo](#) object to another.

The two are 'equal' if they refer to the same mass body.

Parameters

<i>rhs</i>	Object to be compared with this object.
------------	---

Definition at line 139 of file `surface_model.hh`.

8.4.3.2 p()

```
JEOD_CLASS_ESTABLISH_FRIENDS2 jeod::FacetStateInfo::p (
    jeod ,
    FacetStateInfo ) [private]
```

8.4.4 Field Documentation

8.4.4.1 mass_body

```
MassBody* jeod::FacetStateInfo::mass_body {}
```

The MassBody object whose state, relative and w.r.t.

the MassBody named in struct_body_name, is being calculated trick_io(**)

Definition at line 109 of file surface_model.hh.

8.4.4.2 mass_state

```
MassPointState jeod::FacetStateInfo::mass_state
```

The resulting relative mass point state between the structural body named in struct_body_name and the MassBody pointed to in this structure's mass_body.

trick_io(**)

Definition at line 103 of file surface_model.hh.

The documentation for this class was generated from the following file:

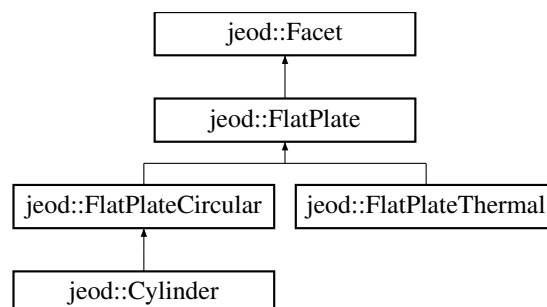
- [surface_model.hh](#)

8.5 jeod::FlatPlate Class Reference

A [FlatPlate](#) implementation of [Facet](#).

```
#include <flat_plate.hh>
```

Inheritance diagram for jeod::FlatPlate:



Public Member Functions

- [FlatPlate](#) ()=default
- [~FlatPlate](#) () override=default
- [FlatPlate](#) & [operator=](#) (const [FlatPlate](#) &)=delete
- [FlatPlate](#) (const [FlatPlate](#) &)=delete

Data Fields

- double [normal](#) [3] {}
normal of the plate, pointing outward of the craft, with respect to the vehicle structural frame.
- double [local_normal](#) [3] {}
The normal of the plate with respect to the structural frame associated with the mass body named in `mass_body_name`.

Protected Member Functions

- void [update_articulation_internal](#) () override

Private Member Functions

- JEOD_CLASS_ESTABLISH_FRIENDS2 [p](#) (jeod, [FlatPlate](#))

Additional Inherited Members

8.5.1 Detailed Description

A [FlatPlate](#) implementation of [Facet](#).

Definition at line 83 of file `flat_plate.hh`.

8.5.2 Constructor & Destructor Documentation

8.5.2.1 FlatPlate() [1/2]

```
jeod::FlatPlate::FlatPlate ( ) [default]
```

8.5.2.2 ~FlatPlate()

```
jeod::FlatPlate::~FlatPlate ( ) [override], [default]
```

8.5.2.3 FlatPlate() [2/2]

```
jeod::FlatPlate::FlatPlate (
    const FlatPlate & ) [delete]
```

8.5.3 Member Function Documentation

8.5.3.1 operator=()

```
FlatPlate& jeod::FlatPlate::operator= (
    const FlatPlate & ) [delete]
```

8.5.3.2 p()

```
JEOD_CLASS_ESTABLISH_FRIENDS2 jeod::FlatPlate::p (
    jeod ,
    FlatPlate ) [private]
```

8.5.3.3 update_articulation_internal()

```
void jeod::FlatPlate::update_articulation_internal ( ) [override], [protected], [virtual]
```

Reimplemented from [jeod::Facet](#).

Definition at line 44 of file flat_plate.cc.

References [local_normal](#), [jeod::Facet::mass_rel_struct](#), [normal](#), and [jeod::Facet::update_articulation_internal\(\)](#).

8.5.4 Field Documentation

8.5.4.1 local_normal

```
double jeod::FlatPlate::local_normal[3] {}
```

The normal of the plate with respect to the structural frame associated with the mass body named in [mass_body↔_name](#).

[trick_units\(-\)](#)

Definition at line 107 of file flat_plate.hh.

Referenced by [update_articulation_internal\(\)](#).

8.5.4.2 normal

```
double jeod::FlatPlate::normal[3] {}
```

normal of the plate, pointing outward of the craft, with respect to the vehicle structural frame.

If the mass tying functionality is turned on, this will be in the structural frame associated with the mass body named in `struct_mass_name`. If mass tying functionality is not turned on, this parameter should be set directly. If it is turned on, then the `local_normal` should be set and this variable should be automatically calculated. `trick_units(-)`

Definition at line 101 of file `flat_plate.hh`.

Referenced by `update_articulation_internal()`.

The documentation for this class was generated from the following files:

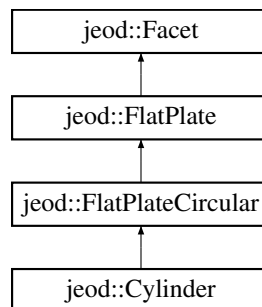
- [flat_plate.hh](#)
- [flat_plate.cc](#)

8.6 jeod::FlatPlateCircular Class Reference

An circular flat plate implementation of [Facet](#).

```
#include <flat_plate_circular.hh>
```

Inheritance diagram for `jeod::FlatPlateCircular`:



Public Member Functions

- [FlatPlateCircular](#) ()=default
- [~FlatPlateCircular](#) () override=default
- [FlatPlateCircular](#) & [operator=](#) (const [FlatPlateCircular](#) &)=delete
- [FlatPlateCircular](#) (const [FlatPlateCircular](#) &)=delete

Data Fields

- double [radius](#) {}
Radius of the plate.

Private Member Functions

- JEOD_CLASS_ESTABLISH_FRIENDS2 [p](#) (jeod, [FlatPlateCircular](#))

Additional Inherited Members

8.6.1 Detailed Description

An circular flat plate implementation of [Facet](#).

Definition at line 74 of file flat_plate_circular.hh.

8.6.2 Constructor & Destructor Documentation

8.6.2.1 FlatPlateCircular() [1/2]

```
jeod::FlatPlateCircular::FlatPlateCircular ( ) [default]
```

8.6.2.2 ~FlatPlateCircular()

```
jeod::FlatPlateCircular::~~FlatPlateCircular ( ) [override], [default]
```

8.6.2.3 FlatPlateCircular() [2/2]

```
jeod::FlatPlateCircular::FlatPlateCircular (
    const FlatPlateCircular & ) [delete]
```

8.6.3 Member Function Documentation

8.6.3.1 operator=()

```
FlatPlateCircular& jeod::FlatPlateCircular::operator= (
    const FlatPlateCircular & ) [delete]
```

8.6.3.2 p()

```
JEOD_CLASS_ESTABLISH_FRIENDS2 jeod::FlatPlateCircular::p (
    jeod ,
    FlatPlateCircular ) [private]
```

8.6.4 Field Documentation

8.6.4.1 radius

```
double jeod::FlatPlateCircular::radius {}
```

Radius of the plate.

trick_units(m)

Definition at line 85 of file flat_plate_circular.hh.

The documentation for this class was generated from the following file:

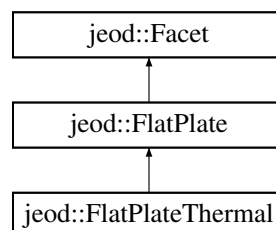
- [flat_plate_circular.hh](#)

8.7 jeod::FlatPlateThermal Class Reference

A [FlatPlate](#) implementation of [Facet](#), with thermal information.

```
#include <flat_plate_thermal.hh>
```

Inheritance diagram for jeod::FlatPlateThermal:



Public Member Functions

- [FlatPlateThermal](#) ()=default
- [~FlatPlateThermal](#) () override=default
- [FlatPlateThermal & operator=](#) (const [FlatPlateThermal](#) &)=delete
- [FlatPlateThermal](#) (const [FlatPlateThermal](#) &)=delete

Data Fields

- ThermalFacetRider [thermal](#)
Thermal characteristics rider.

Private Member Functions

- [JEOD_CLASS_ESTABLISH_FRIENDS2p](#) (jeod, [FlatPlateThermal](#))

Additional Inherited Members

8.7.1 Detailed Description

A [FlatPlate](#) implementation of [Facet](#), with thermal information.

Definition at line 80 of file flat_plate_thermal.hh.

8.7.2 Constructor & Destructor Documentation

8.7.2.1 FlatPlateThermal() [1/2]

```
jeod::FlatPlateThermal::FlatPlateThermal ( ) [default]
```

8.7.2.2 ~FlatPlateThermal()

```
jeod::FlatPlateThermal::~~FlatPlateThermal ( ) [override], [default]
```

8.7.2.3 FlatPlateThermal() [2/2]

```
jeod::FlatPlateThermal::FlatPlateThermal (
    const FlatPlateThermal & ) [delete]
```

8.7.3 Member Function Documentation

8.7.3.1 JEOD_CLASS_ESTABLISH_FRIENDS2p()

```
jeod::FlatPlateThermal::JEOD_CLASS_ESTABLISH_FRIENDS2p (
    jeod ,
    FlatPlateThermal ) [private]
```

8.7.3.2 operator=()

```
FlatPlateThermal& jeod::FlatPlateThermal::operator= (
    const FlatPlateThermal & ) [delete]
```

8.7.4 Field Documentation

8.7.4.1 thermal

```
ThermalFacetRider jeod::FlatPlateThermal::thermal
```

Thermal characteristics rider.

trick_units(-)

Definition at line 91 of file flat_plate_thermal.hh.

The documentation for this class was generated from the following file:

- [flat_plate_thermal.hh](#)

8.8 jeod::InteractionFacet Class Reference

A base class for an interaction specific facet.

```
#include <interaction_facet.hh>
```

Public Member Functions

- [InteractionFacet](#) ()=default
- virtual [~InteractionFacet](#) ()=default
- [InteractionFacet](#) & [operator=](#) (const [InteractionFacet](#) &)=delete
- [InteractionFacet](#) (const [InteractionFacet](#) &)=delete

Data Fields

- double [force](#) [3] {}
The force on the facet caused by the environment interaction.
- double [torque](#) [3] {}
The torque on the facet caused by the environment interaction.
- [Facet](#) * [base_facet](#) {}
The original facet from which this interaction facet was created.

Private Member Functions

- JEOD_CLASS_ESTABLISH_FRIENDS2 [p](#) (jeod, [InteractionFacet](#))

8.8.1 Detailed Description

A base class for an interaction specific facet.

Definition at line 80 of file `interaction_facet.hh`.

8.8.2 Constructor & Destructor Documentation

8.8.2.1 InteractionFacet() [1/2]

```
jeod::InteractionFacet::InteractionFacet ( ) [default]
```

8.8.2.2 ~InteractionFacet()

```
virtual jeod::InteractionFacet::~~InteractionFacet ( ) [virtual], [default]
```

8.8.2.3 InteractionFacet() [2/2]

```
jeod::InteractionFacet::InteractionFacet (
    const InteractionFacet & ) [delete]
```

8.8.3 Member Function Documentation

8.8.3.1 operator=()

```
InteractionFacet& jeod::InteractionFacet::operator= (
    const InteractionFacet & ) [delete]
```

8.8.3.2 p()

```
JEOD_CLASS_ESTABLISH_FRIENDS2 jeod::InteractionFacet::p (
    jeod ,
    InteractionFacet ) [private]
```

8.8.4 Field Documentation

8.8.4.1 base_facet

```
Facet* jeod::InteractionFacet::base_facet {}
```

The original facet from which this interaction facet was created.

trick_units(-)

Definition at line 101 of file interaction_facet.hh.

8.8.4.2 force

```
double jeod::InteractionFacet::force[3] {}
```

The force on the facet caused by the environment interaction.

trick_units(N)

Definition at line 91 of file interaction_facet.hh.

8.8.4.3 torque

```
double jeod::InteractionFacet::torque[3] {}
```

The torque on the facet caused by the environment interaction.

trick_units(N*m)

Definition at line 96 of file interaction_facet.hh.

The documentation for this class was generated from the following file:

- [interaction_facet.hh](#)

8.9 jeod::InteractionFacetFactory Class Reference

A factory to create a specific interaction facet from a general facet.

```
#include <interaction_facet_factory.hh>
```

Public Member Functions

- [InteractionFacetFactory](#) ()=default
- virtual [~InteractionFacetFactory](#) ()=default
- [InteractionFacetFactory](#) & [operator=](#) (const [InteractionFacetFactory](#) &)=delete
- [InteractionFacetFactory](#) (const [InteractionFacetFactory](#) &)=delete
- virtual [InteractionFacet](#) * [create_facet](#) ([Facet](#) *facet, [FacetParams](#) *params)=0
A pure virtual function that creates a specific interaction facet from a base facet with the given [FacetParams](#).
- virtual bool [is_correct_factory](#) ([Facet](#) *facet)=0
A pure virtual function.

Protected Attributes

- bool [trick_bool](#) {}
Unused data field to expedite dynamic allocation in Trick environment.

Private Member Functions

- JEOD_CLASS_ESTABLISH_FRIENDS2 [p](#) (jeod, [InteractionFacetFactory](#))

8.9.1 Detailed Description

A factory to create a specific interaction facet from a general facet.

Definition at line 85 of file `interaction_facet_factory.hh`.

8.9.2 Constructor & Destructor Documentation

8.9.2.1 InteractionFacetFactory() [1/2]

```
jeod::InteractionFacetFactory::InteractionFacetFactory ( ) [default]
```

8.9.2.2 ~InteractionFacetFactory()

```
virtual jeod::InteractionFacetFactory::~~InteractionFacetFactory ( ) [virtual], [default]
```

8.9.2.3 InteractionFacetFactory() [2/2]

```
jeod::InteractionFacetFactory::InteractionFacetFactory (
    const InteractionFacetFactory & ) [delete]
```

8.9.3 Member Function Documentation

8.9.3.1 create_facet()

```
virtual InteractionFacet* jeod::InteractionFacetFactory::create_facet (
    Facet * facet,
    FacetParams * params ) [pure virtual]
```

A pure virtual function that creates a specific interaction facet from a base facet with the given [FacetParams](#).

This defines interface for all classes that inherit from [InteractionFacetFactory](#)

Returns

The new interaction facet

Parameters

in	<i>facet</i>	The facet the InteractionFacet is created from
in	<i>params</i>	The parameter object to be added.

8.9.3.2 is_correct_factory()

```
virtual bool jeod::InteractionFacetFactory::is_correct_factory (
    Facet * facet ) [pure virtual]
```

A pure virtual function.

Returns true or false: is the given facet the type this factory is meant to use?

Returns

true or false. Is the given facet the correct type for this factory?

Parameters

in	<i>facet</i>	The facet that is being checked
----	--------------	---------------------------------

8.9.3.3 operator=()

```
InteractionFacetFactory& jeod::InteractionFacetFactory::operator= (
    const InteractionFacetFactory & ) [delete]
```

8.9.3.4 p()

```
JEOD_CLASS_ESTABLISH_FRIENDS2 jeod::InteractionFacetFactory::p (
    jeod ,
    InteractionFacetFactory ) [private]
```

8.9.4 Field Documentation

8.9.4.1 trick_bool

```
bool jeod::InteractionFacetFactory::trick_bool {} [protected]
```

Unused data field to expedite dynamic allocation in Trick environment.

trick_units(—)

Definition at line 119 of file interaction_facet_factory.hh.

The documentation for this class was generated from the following file:

- [interaction_facet_factory.hh](#)

8.10 jeod::InteractionSurface Class Reference

A base class for interaction specific surfaces.

```
#include <interaction_surface.hh>
```

Public Member Functions

- [InteractionSurface](#) ()=default
- virtual [~InteractionSurface](#) ()=default
- [InteractionSurface](#) & [operator=](#) (const [InteractionSurface](#) &)=delete
- [InteractionSurface](#) (const [InteractionSurface](#) &)=delete
- virtual void [accumulate_thermal_sources](#) ()
Adds all thermal sources together.
- virtual void [thermal_integrator](#) ()
Integrates thermal sources to get temperature.
- virtual void [allocate_array](#) (unsigned int size)=0
A pure virtual function that will allocate the array of pointers to the correct interaction facet type, of the given size.
- virtual void [allocate_interaction_facet](#) ([Facet](#) *facet, [InteractionFacetFactory](#) *factory, [FacetParams](#) *params, unsigned int index)=0
A pure virtual function that will allocate the interaction facet, from the given facet, using the given facet parameters, and place it in the allocated array of interaction facets at the given index.

Private Member Functions

- JEOD_CLASS_ESTABLISH_FRIENDS2 [p](#) (jeod, [InteractionSurface](#))

8.10.1 Detailed Description

A base class for interaction specific surfaces.

Definition at line 84 of file [interaction_surface.hh](#).

8.10.2 Constructor & Destructor Documentation

8.10.2.1 [InteractionSurface](#)() [1/2]

```
jeod::InteractionSurface::InteractionSurface ( ) [default]
```

8.10.2.2 [~InteractionSurface](#)()

```
virtual jeod::InteractionSurface::~~InteractionSurface ( ) [virtual], [default]
```

8.10.2.3 [InteractionSurface](#)() [2/2]

```
jeod::InteractionSurface::InteractionSurface (
    const InteractionSurface & ) [delete]
```

8.10.3 Member Function Documentation

8.10.3.1 accumulate_thermal_sources()

```
virtual void jeod::InteractionSurface::accumulate_thermal_sources ( ) [inline], [virtual]
```

Adds all thermal sources together.

Definition at line 100 of file `interaction_surface.hh`.

8.10.3.2 allocate_array()

```
virtual void jeod::InteractionSurface::allocate_array (
    unsigned int size ) [pure virtual]
```

A pure virtual function that will allocate the array of pointers to the correct interaction facet type, of the given size.

Parameters

in	<i>size</i>	Size of the array to be allocated Units: cnt
----	-------------	---

Referenced by `jeod::InteractionSurfaceFactory::create_surface()`.

8.10.3.3 allocate_interaction_facet()

```
virtual void jeod::InteractionSurface::allocate_interaction_facet (
    Facet * facet,
    InteractionFacetFactory * factory,
    FacetParams * params,
    unsigned int index ) [pure virtual]
```

A pure virtual function that will allocate the interaction facet, from the given facet, using the given facet parameters, and place it in the allocated array of interaction facets at the given index.

Parameters

in	<i>facet</i>	The facet used to create the interaction facet
in	<i>factory</i>	The factory used to create the interaction facet
in	<i>params</i>	The parameters used to create the interaction facet
in	<i>index</i>	Where in the interaction facet array the interaction facet will be placed Units: cnt

Referenced by `jeod::InteractionSurfaceFactory::create_surface()`.

8.10.3.4 operator=()

```
InteractionSurface& jeod::InteractionSurface::operator= (
    const InteractionSurface & ) [delete]
```

8.10.3.5 p()

```
JEOD_CLASS_ESTABLISH_FRIENDS2 jeod::InteractionSurface::p (
    jeod ,
    InteractionSurface ) [private]
```

8.10.3.6 thermal_integrator()

```
virtual void jeod::InteractionSurface::thermal_integrator ( ) [inline], [virtual]
```

Integrates thermal sources to get temperature.

Definition at line 110 of file `interaction_surface.hh`.

The documentation for this class was generated from the following file:

- [interaction_surface.hh](#)

8.11 jeod::InteractionSurfaceFactory Class Reference

A base class for creating specific interaction surfaces from general surfaces.

```
#include <interaction_surface_factory.hh>
```

Public Member Functions

- [InteractionSurfaceFactory](#) ()
Default Constructor.
- virtual [~InteractionSurfaceFactory](#) ()
Destructor.
- [InteractionSurfaceFactory](#) & operator= (const [InteractionSurfaceFactory](#) &)=delete
- [InteractionSurfaceFactory](#) (const [InteractionSurfaceFactory](#) &)=delete
- virtual void [create_surface](#) ([SurfaceModel](#) *surface, [InteractionSurface](#) *inter_surface)
Creates an interaction surface, in the inter_surface parameter, from the given SurfaceModel.
- void [create_surface](#) ([SurfaceModel](#) &surface, [InteractionSurface](#) &inter_surface)
Convenience version of create_surface which can be called from the input file.
- virtual void [add_facet_factory](#) ([InteractionFacetFactory](#) *to_add)
Used to add an interaction facet factory for use in the surface factory.
- virtual void [add_facet_params](#) ([FacetParams](#) *to_add)
Add a set of facet parameters for use in the interaction surface factory.

Data Fields

- JeodPointerVector< [InteractionFacetFactory](#) >::type [factories](#)
A vector of interaction facet factories to be used.
- JeodPointerVector< [FacetParams](#) >::type [params](#)
A vector of [FacetParams](#) to be used.

Private Member Functions

- JEOD_CLASS_ESTABLISH_FRIENDS2 [p](#) (jeod, [InteractionSurfaceFactory](#))

8.11.1 Detailed Description

A base class for creating specific interaction surfaces from general surfaces.

Definition at line 88 of file `interaction_surface_factory.hh`.

8.11.2 Constructor & Destructor Documentation

8.11.2.1 InteractionSurfaceFactory() [1/2]

```
jeod::InteractionSurfaceFactory::InteractionSurfaceFactory ( )
```

Default Constructor.

Definition at line 58 of file `interaction_surface_factory.cc`.

References [factories](#), and [params](#).

8.11.2.2 ~InteractionSurfaceFactory()

```
jeod::InteractionSurfaceFactory::~~InteractionSurfaceFactory ( ) [virtual]
```

Destructor.

Definition at line 71 of file `interaction_surface_factory.cc`.

References [factories](#), and [params](#).

8.11.2.3 InteractionSurfaceFactory() [2/2]

```
jeod::InteractionSurfaceFactory::InteractionSurfaceFactory (
    const InteractionSurfaceFactory & ) [delete]
```

8.11.3 Member Function Documentation

8.11.3.1 add_facet_factory()

```
void jeod::InteractionSurfaceFactory::add_facet_factory (
    InteractionFacetFactory * to_add ) [virtual]
```

Used to add an interaction facet factory for use in the surface factory.

Parameters

<i>in</i>	<i>to_add</i>	The interaction facet factory to add
-----------	---------------	--------------------------------------

Definition at line 183 of file `interaction_surface_factory.cc`.

References `factories`, and `jeod::SurfaceModelMessages::setup_error`.

8.11.3.2 add_facet_params()

```
void jeod::InteractionSurfaceFactory::add_facet_params (
    FacetParams * to_add ) [virtual]
```

Add a set of facet parameters for use in the interaction surface factory.

Parameters

<i>in</i>	<i>to_add</i>	The facet parameters to add
-----------	---------------	-----------------------------

Definition at line 203 of file `interaction_surface_factory.cc`.

References `jeod::FacetParams::name`, `params`, and `jeod::SurfaceModelMessages::setup_error`.

8.11.3.3 create_surface() [1/2]

```
void jeod::InteractionSurfaceFactory::create_surface (
    SurfaceModel & surface,
    InteractionSurface & inter_surface ) [inline]
```

Convenience version of `create_surface` which can be called from the input file.

Parameters

<i>surface</i>	Surface model.
<i>inter_surface</i>	Reference to the interaction surface.

Definition at line 105 of file `interaction_surface_factory.hh`.

8.11.3.4 create_surface() [2/2]

```
void jeod::InteractionSurfaceFactory::create_surface (
    SurfaceModel * surface,
    InteractionSurface * inter_surface ) [virtual]
```

Creates an interaction surface, in the `inter_surface` parameter, from the given [SurfaceModel](#).

The [InteractionSurfaceFactory](#) should contain all necessary [InteractionFacetFactories](#) and [FacetParams](#) already

Parameters

in	<i>surface</i>	The surface model used to create the interaction surface
out	<i>inter_surface</i>	Where the interaction surface will be produced

Definition at line 86 of file `interaction_surface_factory.cc`.

References `jeod::InteractionSurface::allocate_array()`, `jeod::InteractionSurface::allocate_interaction_facet()`, `jeod::SurfaceModel::facets`, `factories`, `jeod::SurfaceModelMessages::initialization_error`, and `params`.

8.11.3.5 operator=()

```
InteractionSurfaceFactory& jeod::InteractionSurfaceFactory::operator= (
    const InteractionSurfaceFactory & ) [delete]
```

8.11.3.6 p()

```
JEOD_CLASS_ESTABLISH_FRIENDS2 jeod::InteractionSurfaceFactory::p (
    jeod ,
    InteractionSurfaceFactory ) [private]
```

8.11.4 Field Documentation

8.11.4.1 factories

```
JeodPointerVector<InteractionFacetFactory>::type jeod::InteractionSurfaceFactory::factories
```

A vector of interaction facet factories to be used.

Matched to facets by type `trick_io(**)`

Definition at line 114 of file `interaction_surface_factory.hh`.

Referenced by `add_facet_factory()`, `create_surface()`, `InteractionSurfaceFactory()`, and `~InteractionSurfaceFactory()`.

8.11.4.2 params

```
JeodPointerVector<FacetParams>::type jeod::InteractionSurfaceFactory::params
```

A vector of [FacetParams](#) to be used.

```
trick_io(**)
```

Definition at line 119 of file `interaction_surface_factory.hh`.

Referenced by `add_facet_params()`, `create_surface()`, `InteractionSurfaceFactory()`, and `~InteractionSurfaceFactory()`.

The documentation for this class was generated from the following files:

- [interaction_surface_factory.hh](#)
- [interaction_surface_factory.cc](#)

8.12 jeod::SurfaceModel Class Reference

A general, non-interaction specific surface that can be used to create surfaces suitable for specific interactions.

```
#include <surface_model.hh>
```

Public Member Functions

- [SurfaceModel](#) ()
Default constructor.
- [~SurfaceModel](#) ()
Destructor.
- [SurfaceModel](#) & `operator=` (const [SurfaceModel](#) &)=delete
- [SurfaceModel](#) (const [SurfaceModel](#) &)=delete
- void [add_facets](#) ([Facet](#) **new_facets, unsigned int num_new_facets)
- void [add_facet](#) ([Facet](#) *new_facet)
- void [initialize_mass_connections](#) (BaseDynManager &manager)
- void [update_articulation](#) ()

Data Fields

- bool [articulation_active](#) {}
Is the articulation active? If yes, facet information will be updated from the previously supplied mass tree.
- std::string [struct_body_name](#)
The name of the MassBody representing the overall structural frame of the vehicle associated with this surface model.
- JeodPointerVector< [Facet](#) >::type [facets](#)
The facets that make up the surface.

Protected Attributes

- MassBody * [struct_body_ptr](#) {}
A pointer to the MassBody named by struct_body_name.
- JeodPointerList< [FacetStateInfo](#) >::type [articulation_states](#)
The set of states used to update the articulation of each facet.

Private Member Functions

- JEOD_CLASS_ESTABLISH_FRIENDS2 [p](#) (jeod, [SurfaceModel](#))

8.12.1 Detailed Description

A general, non-interaction specific surface that can be used to create surfaces suitable for specific interactions.

Definition at line 149 of file `surface_model.hh`.

8.12.2 Constructor & Destructor Documentation

8.12.2.1 SurfaceModel() [1/2]

```
jeod::SurfaceModel::SurfaceModel ( )
```

Default constructor.

Definition at line 56 of file `surface_model.cc`.

References `articulation_states`, and `facets`.

8.12.2.2 ~SurfaceModel()

```
jeod::SurfaceModel::~~SurfaceModel ( )
```

Destructor.

Definition at line 69 of file `surface_model.cc`.

References `articulation_states`, and `facets`.

8.12.2.3 SurfaceModel() [2/2]

```
jeod::SurfaceModel::SurfaceModel (
    const SurfaceModel & ) [delete]
```

8.12.3 Member Function Documentation

8.12.3.1 add_facet()

```
void jeod::SurfaceModel::add_facet (
    Facet * new_facet )
```

Definition at line 121 of file surface_model.cc.

References facets, and jeod::SurfaceModelMessages::setup_error.

8.12.3.2 add_facets()

```
void jeod::SurfaceModel::add_facets (
    Facet ** new_facets,
    unsigned int num_new_facets )
```

Definition at line 87 of file surface_model.cc.

References facets, and jeod::SurfaceModelMessages::setup_error.

8.12.3.3 initialize_mass_connections()

```
void jeod::SurfaceModel::initialize_mass_connections (
    BaseDynManager & manager )
```

Definition at line 143 of file surface_model.cc.

References articulation_states, facets, jeod::SurfaceModelMessages::initialization_error, struct_body_name, and struct_body_ptr.

8.12.3.4 operator=()

```
SurfaceModel& jeod::SurfaceModel::operator= (
    const SurfaceModel & ) [delete]
```

8.12.3.5 p()

```
JEOD_CLASS_ESTABLISH_FRIENDS2 jeod::SurfaceModel::p (
    jeod ,
    SurfaceModel ) [private]
```

8.12.3.6 update_articulation()

```
void jeod::SurfaceModel::update_articulation ( )
```

Definition at line 202 of file surface_model.cc.

References articulation_active, articulation_states, facets, jeod::SurfaceModelMessages::runtime_error, struct_↔ body_name, and struct_body_ptr.

8.12.4 Field Documentation

8.12.4.1 articulation_active

```
bool jeod::SurfaceModel::articulation_active {}
```

Is the articulation active? If yes, facet information will be updated from the previously supplied mass tree.

If not, nothing will update. This defaults to false trick_units(-)

Definition at line 162 of file surface_model.hh.

Referenced by update_articulation().

8.12.4.2 articulation_states

```
JeodPointerList<FacetStateInfo>::type jeod::SurfaceModel::articulation_states [protected]
```

The set of states used to update the articulation of each facet.

```
trick_io(**)
```

Definition at line 201 of file surface_model.hh.

Referenced by initialize_mass_connections(), SurfaceModel(), update_articulation(), and ~SurfaceModel().

8.12.4.3 facets

```
JeodPointerVector<Facet>::type jeod::SurfaceModel::facets
```

The facets that make up the surface.

```
trick_io(**)
```

Definition at line 185 of file `surface_model.hh`.

Referenced by `add_facet()`, `add_facets()`, `jeod::InteractionSurfaceFactory::create_surface()`, `initialize_mass_connections()`, `SurfaceModel()`, `update_articulation()`, and `~SurfaceModel()`.

8.12.4.4 struct_body_name

```
std::string jeod::SurfaceModel::struct_body_name
```

The name of the `MassBody` representing the overall structural frame of the vehicle associated with this surface model.

All states of all contained facets will be relative to the structural frame of this `MassBody`. This name is only required for specific applications, such as contact and articulation `trick_units(-)`

Definition at line 180 of file `surface_model.hh`.

Referenced by `initialize_mass_connections()`, and `update_articulation()`.

8.12.4.5 struct_body_ptr

```
MassBody* jeod::SurfaceModel::struct_body_ptr {} [protected]
```

A pointer to the `MassBody` named by `struct_body_name`.

This pointer will be set, using `struct_body_name`, by searching the `DynManager` object supplied to the `initialize_mass_connections()` function. This pointer is only used for specific applications, such as contact and articulation `trick_units(-)`

Definition at line 195 of file `surface_model.hh`.

Referenced by `initialize_mass_connections()`, and `update_articulation()`.

The documentation for this class was generated from the following files:

- [surface_model.hh](#)
- [surface_model.cc](#)

8.13 jeod::SurfaceModelMessages Class Reference

Messages associated with the use of the surface model.

```
#include <surface_model_messages.hh>
```

Public Member Functions

- [SurfaceModelMessages](#) ()=delete
- [SurfaceModelMessages](#) (const [SurfaceModelMessages](#) &rhs)=delete
- [SurfaceModelMessages](#) & operator= (const [SurfaceModelMessages](#) &rhs)=delete

Static Public Attributes

- static const char * [initialization_error](#) = "utils/surface_model/" "initialization_error"
Represents an error in initialization.
- static const char * [setup_error](#) = "utils/surface_model/" "setup_error"
Represents an error in setup of the surface model.
- static const char * [runtime_error](#) = "utils/surface_model/" "runtime_error"
Represents an error during the runtime of the surface model.

Private Member Functions

- [JEOD_CLASS_ESTABLISH_FRIENDS2](#) (jeod, [SurfaceModelMessages](#))

8.13.1 Detailed Description

Messages associated with the use of the surface model.

Definition at line 89 of file `surface_model_messages.hh`.

8.13.2 Constructor & Destructor Documentation

8.13.2.1 SurfaceModelMessages() [1/2]

```
jeod::SurfaceModelMessages::SurfaceModelMessages ( ) [delete]
```

8.13.2.2 SurfaceModelMessages() [2/2]

```
jeod::SurfaceModelMessages::SurfaceModelMessages (
    const SurfaceModelMessages & rhs ) [delete]
```

8.13.3 Member Function Documentation

8.13.3.1 JEOD_CLASS_ESTABLISH_FRIENDS2()

```
jeod::SurfaceModelMessages::JEOD_CLASS_ESTABLISH_FRIENDS2 (
    jeod ,
    SurfaceModelMessages ) [private]
```

8.13.3.2 operator=()

```
SurfaceModelMessages& jeod::SurfaceModelMessages::operator= (
    const SurfaceModelMessages & rhs ) [delete]
```

8.13.4 Field Documentation

8.13.4.1 initialization_error

```
char const * jeod::SurfaceModelMessages::initialization_error = "utils/surface_model/" "initialization_↵
_error" [static]
```

Represents an error in initialization.

trick_units(−)

Definition at line 99 of file surface_model_messages.hh.

Referenced by jeod::InteractionSurfaceFactory::create_surface(), jeod::Facet::initialize_mass_connection(), jeod↵
::SurfaceModel::initialize_mass_connections(), and jeod::Facet::update_articulation().

8.13.4.2 runtime_error

```
char const * jeod::SurfaceModelMessages::runtime_error = "utils/surface_model/" "runtime_↵
error" [static]
```

Represents an error during the runtime of the surface model.

trick_units(−)

Definition at line 109 of file surface_model_messages.hh.

Referenced by jeod::SurfaceModel::update_articulation().

8.13.4.3 setup_error

```
char const * jeod::SurfaceModelMessages::setup_error = "utils/surface_model/" "setup_error"  
[static]
```

Represents an error in setup of the surface model.

trick_units(−)

Definition at line 104 of file surface_model_messages.hh.

Referenced by `jeod::SurfaceModel::add_facet()`, `jeod::InteractionSurfaceFactory::add_facet_factory()`, `jeod::InteractionSurfaceFactory::add_facet_params()`, and `jeod::SurfaceModel::add_facets()`.

The documentation for this class was generated from the following files:

- [surface_model_messages.hh](#)
- [surface_model_messages.cc](#)

Chapter 9

File Documentation

9.1 `class_declarations.hh` File Reference

Forward declarations of classes defined for JEOD 2.0 surface model.

Namespaces

- [jeod](#)

Namespace jeod.

9.1.1 Detailed Description

Forward declarations of classes defined for JEOD 2.0 surface model.

9.2 `cylinder.hh` File Reference

cylinders for use in the surface model and the contact model

```
#include "flat_plate_circular.hh"
#include "utils/sim_interface/include/jeod_class.hh"
```

Data Structures

- class [jeod::Cylinder](#)

An cylinder implementation of [Facet](#).

Namespaces

- [jeod](#)

Namespace jeod.

9.2.1 Detailed Description

cylinders for use in the surface model and the contact model

9.3 facet.cc File Reference

Individual facets for use in the surface model.

```
#include <cstdint>
#include "dynamics/dyn_manager/include/base_dyn_manager.hh"
#include "dynamics/mass/include/mass.hh"
#include "utils/math/include/vector3.hh"
#include "utils/message/include/message_handler.hh"
#include "utils/named_item/include/named_item.hh"
#include "../include/facet.hh"
#include "../include/surface_model_messages.hh"
```

Namespaces

- [jeod](#)

Namespace jeod.

9.3.1 Detailed Description

Individual facets for use in the surface model.

9.4 facet.hh File Reference

Individual facets for use in the surface model.

```
#include <string>
#include <utility>
#include "utils/sim_interface/include/jeod_class.hh"
#include "dynamics/mass/include/mass_point_state.hh"
```

Data Structures

- class [jeod::Facet](#)

A general base class for all surface model facets.

Namespaces

- [jeod](#)

Namespace jeod.

9.4.1 Detailed Description

Individual facets for use in the surface model.

9.5 facet_params.hh File Reference

A virtual base class for facet parameters, used to create interaction facets in the InteractionSurfaceFactorys.

```
#include <string>
#include <utility>
#include "utils/sim_interface/include/jeod_class.hh"
```

Data Structures

- class [jeod::FacetParams](#)

General base class for all parameters associated with facets in the surface model.

Namespaces

- [jeod](#)

Namespace jeod.

9.5.1 Detailed Description

A virtual base class for facet parameters, used to create interaction facets in the InteractionSurfaceFactorys.

9.6 flat_plate.cc File Reference

Flat plates for use in the surface model.

```
#include "../include/flat_plate.hh"
#include "utils/math/include/vector3.hh"
```

Namespaces

- [jeod](#)

Namespace jeod.

9.6.1 Detailed Description

Flat plates for use in the surface model.

9.7 flat_plate.hh File Reference

Flat plates for use in the surface model.

```
#include "utils/sim_interface/include/jeod_class.hh"
#include "facet.hh"
```

Data Structures

- class [jeod::FlatPlate](#)
A [FlatPlate](#) implementation of [Facet](#).

Namespaces

- [jeod](#)
Namespace [jeod](#).

9.7.1 Detailed Description

Flat plates for use in the surface model.

9.8 flat_plate_circular.hh File Reference

circulat flat plates for use in the surface model and the contact model

```
#include "flat_plate.hh"
#include "utils/sim_interface/include/jeod_class.hh"
```

Data Structures

- class [jeod::FlatPlateCircular](#)
An circular flat plate implementation of [Facet](#).

Namespaces

- [jeod](#)
Namespace [jeod](#).

9.8.1 Detailed Description

circulat flat plates for use in the surface model and the contact model

9.9 flat_plate_thermal.hh File Reference

Flat plates for use in the surface model, including a thermal portion.

```
#include "interactions/thermal_rider/include/thermal_facet_rider.hh"
#include "utils/sim_interface/include/jeod_class.hh"
#include "flat_plate.hh"
```

Data Structures

- class [jeod::FlatPlateThermal](#)
A [FlatPlate](#) implementation of [Facet](#), with thermal information.

Namespaces

- [jeod](#)
Namespace jeod.

9.9.1 Detailed Description

Flat plates for use in the surface model, including a thermal portion.

9.10 interaction_facet.hh File Reference

Individual facets for use with specific environment interaction models.

```
#include "utils/sim_interface/include/jeod_class.hh"
```

Data Structures

- class [jeod::InteractionFacet](#)
A base class for an interaction specific facet.

Namespaces

- [jeod](#)
Namespace jeod.

9.10.1 Detailed Description

Individual facets for use with specific environment interaction models.

9.11 interaction_facet_factory.hh File Reference

Factory that creates an interaction facet, for a specific environment interaction model, from a facet model.

```
#include "utils/sim_interface/include/jeod_class.hh"
```

Data Structures

- class [jeod::InteractionFacetFactory](#)
A factory to create a specific interaction facet from a general facet.

Namespaces

- [jeod](#)
Namespace jeod.

9.11.1 Detailed Description

Factory that creates an interaction facet, for a specific environment interaction model, from a facet model.

This is a pure virtual class, and the pure virtual functions must be implemented in any instantiable, inheriting class

9.12 interaction_surface.hh File Reference

Vehicle surface model for general environment interaction models.

```
#include "utils/sim_interface/include/jeod_class.hh"
```

Data Structures

- class [jeod::InteractionSurface](#)
A base class for interaction specific surfaces.

Namespaces

- [jeod](#)
Namespace jeod.

9.12.1 Detailed Description

Vehicle surface model for general environment interaction models.

This is a pure virtual function and has methods that must be implemented in any inherited, instantiable class

9.13 interaction_surface_factory.cc File Reference

Factory that creates an interaction surface, for a specific environment interaction model, from a surface model.

```
#include <cstdint>
#include "utils/memory/include/jeod_alloc.hh"
#include "utils/message/include/message_handler.hh"
#include "utils/named_item/include/named_item.hh"
#include "../include/facet.hh"
#include "../include/facet_params.hh"
#include "../include/interaction_facet_factory.hh"
#include "../include/interaction_surface.hh"
#include "../include/interaction_surface_factory.hh"
#include "../include/surface_model.hh"
#include "../include/surface_model_messages.hh"
```

Namespaces

- [jeod](#)

Namespace jeod.

9.13.1 Detailed Description

Factory that creates an interaction surface, for a specific environment interaction model, from a surface model.

9.14 interaction_surface_factory.hh File Reference

Factory that creates an interaction surface, for a specific environment interaction model, from a surface model.

```
#include "utils/container/include/pointer_vector.hh"
#include "utils/sim_interface/include/jeod_class.hh"
```

Data Structures

- class [jeod::InteractionSurfaceFactory](#)

A base class for creating specific interaction surfaces from general surfaces.

Namespaces

- [jeod](#)

Namespace jeod.

9.14.1 Detailed Description

Factory that creates an interaction surface, for a specific environment interaction model, from a surface model.

9.15 surface_model.cc File Reference

Vehicle surface model for general environment interaction models.

```
#include <algorithm>
#include <cstdlib>
#include "dynamics/dyn_manager/include/base_dyn_manager.hh"
#include "dynamics/mass/include/mass.hh"
#include "utils/memory/include/jeod_alloc.hh"
#include "utils/message/include/message_handler.hh"
#include "utils/named_item/include/named_item.hh"
#include "../include/facet.hh"
#include "../include/surface_model.hh"
#include "../include/surface_model_messages.hh"
```

Namespaces

- [jeod](#)

Namespace jeod.

9.15.1 Detailed Description

Vehicle surface model for general environment interaction models.

9.16 surface_model.hh File Reference

Vehicle surface model for general environment interaction models.

```
#include "dynamics/mass/include/mass_point_state.hh"
#include "utils/container/include/object_list.hh"
#include "utils/container/include/object_vector.hh"
#include "utils/container/include/pointer_list.hh"
#include "utils/container/include/pointer_vector.hh"
#include "utils/sim_interface/include/jeod_class.hh"
```

Data Structures

- class [jeod::FacetStateInfo](#)

This is a structure used only in the surface model to aid in relative state calculations for articulation.

- class [jeod::SurfaceModel](#)

A general, non-interaction specific surface that can be used to create surfaces suitable for specific interactions.

Namespaces

- [jeod](#)

Namespace jeod.

9.16.1 Detailed Description

Vehicle surface model for general environment interaction models.

9.17 surface_model_messages.cc File Reference

Implement surface_model_messages.

```
#include "utils/message/include/make_message_code.hh"
#include "../include/surface_model_messages.hh"
```

Namespaces

- [jeod](#)
Namespace jeod.

Macros

- #define [MAKE_SURFACEMODEL_MESSAGE_CODE](#)(id) JEOD_MAKE_MESSAGE_CODE(SurfaceModelMessages, "utils/surface_model/", id)

9.17.1 Detailed Description

Implement surface_model_messages.

9.17.2 Macro Definition Documentation

9.17.2.1 MAKE_SURFACEMODEL_MESSAGE_CODE

```
#define MAKE_SURFACEMODEL_MESSAGE_CODE(  
    id ) JEOD_MAKE_MESSAGE_CODE(SurfaceModelMessages, "utils/surface_model/", id)
```

Definition at line 43 of file surface_model_messages.cc.

9.18 surface_model_messages.hh File Reference

Implement surface_model_messages.

```
#include "utils/sim_interface/include/jeod_class.hh"
#include "class_declarations.hh"
```

Data Structures

- class [jeod::SurfaceModelMessages](#)
Messages associated with the use of the surface model.

Namespaces

- [jeod](#)
Namespace jeod.

9.18.1 Detailed Description

Implement surface_model_messages.

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